SALES PREDICTION USING PYTHON

PROJECT REPORT

Submitted in partial fulfillment of the requirements for the project

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Project Technology:
Python | Pandas | Matplotlib | Seaborn | Scikit-learn

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ABSTRACT

Sales prediction is an important application of data analysis and machine learning. It helps businesses understand the relationship between advertising expenditure and sales performance.

The main objective of this project is to develop a sales prediction model using Python and the Linear Regression algorithm. The project uses an advertising dataset containing information about expenditure on TV, Radio, and Newspaper advertising, along with the corresponding sales figures.

The dataset is analyzed using Pandas and visualized with Matplotlib and Seaborn. The data is divided into training and testing sets, and a Linear Regression model is trained to predict sales. The performance of the model is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared score.

The project demonstrates how machine learning can be used to analyze advertising data and estimate sales based on advertising expenditure.

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1. INTRODUCTION

In today's competitive business environment, predicting sales is important for planning, budgeting, marketing, and decision-making. Businesses invest money in different advertising channels to promote their products and services. Understanding how advertising expenditure relates to sales can help businesses make better decisions.

Sales prediction is the process of estimating future or expected sales using available data and statistical or machine learning techniques. Python provides several libraries that make data analysis and prediction easier.

This project, titled “Sales Prediction Using Python,” uses a Linear Regression model to estimate sales based on advertising expenditure. The dataset contains three advertising features: TV, Radio, and Newspaper. Sales is the target variable that the model attempts to predict.

The project begins with loading and exploring the dataset. Missing values and data types are checked, and graphs are created to understand the relationships between variables. The dataset is then divided into training and testing sets. A Linear Regression model is trained on the training data and evaluated on unseen test data.

The project provides practical experience in Python programming, data visualization, regression analysis, and machine learning model evaluation.

2. OBJECTIVES OF THE PROJECT

The main objective of this project is to develop a machine learning model that predicts sales based on advertising expenditure.

The specific objectives are:

1. To understand the relationship between advertising expenditure and sales.
2. To load and analyze the advertising dataset using Python.
3. To perform exploratory data analysis on the available data.
4. To visualize the relationships between advertising channels and sales.
5. To implement the Linear Regression algorithm.
6. To divide the dataset into training and testing sets.
7. To predict sales using the trained machine learning model.
8. To evaluate model performance using MAE, MSE, RMSE, and R-squared score.
9. To understand the practical application of machine learning in business decision-making.

The project aims to demonstrate how data analysis and regression techniques can help estimate sales from advertising expenditure.

3. PROBLEM STATEMENT

Businesses spend money on different advertising channels, including television, radio, and newspapers. However, understanding the relationship between advertising expenditure and sales can be challenging without proper data analysis.

The problem addressed in this project is to estimate sales using the available advertising expenditure data. A machine learning model is developed to identify relationships between the input variables and the target variable.

The project uses TV, Radio, and Newspaper advertising expenditure as input features and Sales as the output variable. Linear Regression is applied to learn these relationships from historical data.

The trained model is then tested on unseen data to evaluate how well it predicts sales.

The purpose is to build a simple and understandable predictive model that demonstrates the usefulness of machine learning in sales analysis.

4. SCOPE OF THE PROJECT

The scope of this project is limited to predicting sales using advertising expenditure data and a Linear Regression model.

The project includes the following activities:

1. Loading the advertising dataset into Python.
2. Examining the dataset structure and statistical properties.
3. Checking for missing values and data quality issues.
4. Visualizing correlations and relationships between variables.
5. Training a Linear Regression model using historical data.
6. Generating predictions for test data and new advertising expenditure values.
7. Evaluating the model using standard regression metrics.
8. Saving the trained model and prediction results.

This project can serve as a foundation for understanding predictive analytics. However, the model's predictions depend on the quality and characteristics of the dataset. Actual business sales may also be influenced by pricing, competition, seasonality, customer preferences, and other factors not included in this dataset.

5. TOOLS AND TECHNOLOGIES USED

The project uses Python and several data science libraries to analyze the dataset and build a predictive model.

1. Python

Python is the primary programming language used to perform data analysis, visualization, model training, and sales prediction.

2. Google Colab

Google Colab is a cloud-based notebook environment used to write and execute Python code without requiring a local Python installation.

3. Pandas

Pandas is used to load the CSV dataset, inspect records, examine missing values, calculate descriptive statistics, and organize data into tables.

4. NumPy

NumPy supports numerical operations and mathematical calculations, including the calculation of Root Mean Squared Error.

5. Matplotlib

Matplotlib is used to create graphs that help visualize advertising expenditure and compare actual sales with predicted sales.

6. Seaborn

Seaborn is used to create statistical visualizations, including a correlation heatmap.

7. Scikit-learn

Scikit-learn provides the Linear Regression algorithm, the train-test splitting function, and regression evaluation metrics.

8. Joblib

Joblib is used to save the trained model in a file so that it can be loaded and reused later.

6. DATASET DESCRIPTION

The dataset used in this project is an advertising dataset that contains information about advertising expenditure and sales. It is used to study the relationship between advertising investment and sales performance.

The dataset contains 200 records and four columns. Three columns represent advertising expenditure, while the fourth column represents sales.

Description of Variables

1. **TV:** Advertising expenditure on television.
2. **Radio:** Advertising expenditure on radio.
3. **Newspaper:** Advertising expenditure on newspaper advertising.
4. **Sales:** The sales value associated with the advertising expenditure.

The TV, Radio, and Newspaper columns are the independent variables, which are used as inputs to the prediction model. Sales is the dependent variable or target that the model predicts.

The dataset is stored in CSV format and is loaded into Python using the Pandas library.

Dataset Characteristics

- File format: CSV
- Number of records: 200
- Number of columns: 4
- Input features: 3
- Target variable: Sales

The dataset is suitable for demonstrating regression analysis because it contains numerical variables that can be used to predict a continuous target value.

7. SYSTEM REQUIREMENTS

The project was developed using Google Colab, which provides an online environment for executing Python code.

Hardware Requirements

- A computer or laptop
- An internet connection
- A keyboard and pointing device
- Sufficient memory to run a small Python notebook

Software Requirements

- Operating system: Windows, Linux, or macOS
- Platform: Google Colab
- Programming language: Python
- Data analysis library: Pandas
- Numerical library: NumPy
- Visualization libraries: Matplotlib and Seaborn
- Machine learning library: Scikit-learn
- Model persistence library: Joblib
- Web browser: Google Chrome or another compatible browser

Internet Requirements

An internet connection is required to access Google Colab and load the dataset from its online source.

The project does not require a powerful computer because the dataset is small and the Linear Regression algorithm is computationally inexpensive.

8. METHODOLOGY

The project follows a systematic process to develop and evaluate a sales prediction model.

1. Data Collection

The advertising dataset is obtained in CSV format. It contains advertising expenditure for TV, Radio, and Newspaper, along with corresponding sales values.

2. Data Loading

The dataset is loaded into a Pandas DataFrame using the read_csv() function.

3. Data Exploration

The first few rows, dataset dimensions, column names, data types, missing values, and statistical summaries are examined to understand the structure of the data.

4. Data Visualization

Graphs and a correlation heatmap are created to explore relationships between advertising expenditure and sales.

5. Feature Selection

TV, Radio, and Newspaper are selected as input features, while Sales is selected as the target variable.

6. Data Splitting

The dataset is divided into training and testing sets. In this project, 80% of the data is used for training and 20% for testing.

7. Model Training

A Linear Regression model is trained on the training dataset to learn the relationship between advertising expenditure and sales.

8. Prediction

The trained model predicts sales values for the test dataset and a new example of advertising expenditure.

9. Model Evaluation

The predictions are evaluated using Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and the R-squared score.

10. Saving the Results

The trained model is saved using Joblib, and the prediction results are exported to a CSV file.

This methodology provides a clear workflow from data collection to model evaluation and saving the project outputs.

9. DATA LOADING AND EXPLORATION

The first step in the project was to load the advertising dataset into Python using the Pandas library. The dataset was loaded from a CSV file and examined to understand its structure and contents.

The dataset contains information about advertising expenditure on TV, Radio, and Newspaper, along with the corresponding Sales values.

The following activities were performed during data exploration:

- Displayed the first five rows of the dataset.
- Checked the number of rows and columns.
- Examined column names and data types.
- Checked for missing values.
- Generated descriptive statistics to understand the distribution of the data.
- Examined the relationship between advertising expenditure and sales.

Data exploration helped in understanding the dataset before building the machine learning model.

10. DATA PREPROCESSING

Data preprocessing is an important step in a machine learning project. It helps ensure that the dataset is suitable for training and testing the model.

In this project, the following preprocessing activities were performed:

1. Feature Selection: The columns TV, Radio, and Newspaper were selected as input features because they represent advertising expenditure.

2. Target Selection: The Sales column was selected as the target variable because the main objective was to predict sales.

3. Missing Value Inspection: The dataset was checked for missing values using Pandas. Any necessary treatment would depend on the results of this check.

4. Splitting the Dataset: The dataset was divided into training and testing sets. Eighty percent of the data was used for training, while the remaining twenty percent was used for testing.

5. Reproducibility: The random state was set to 42 during the train-test split so that the same split could be reproduced when the code was run again.

After these steps, the data was ready for training the Linear Regression model.

11. DATA VISUALIZATION

Data visualization is the graphical representation of data. It helps in understanding patterns, trends, and relationships between variables.

In this project, Matplotlib and Seaborn libraries were used to visualize the advertising dataset.

The following visualizations were created:

1. Correlation Heatmap

A correlation heatmap was created to examine the relationships between TV, Radio, Newspaper advertising expenditure, and Sales. It helps identify which advertising channels have stronger relationships with sales.

2. TV Advertising vs Sales Scatter Plot

A scatter plot was created to visualize the relationship between TV advertising expenditure and sales. Each point represents an observation from the dataset.

3. Actual vs Predicted Sales Plot

After training the model, a scatter plot was created to compare actual sales values with predicted sales values. A reference line was included to help assess how closely the predictions matched the actual values.

These visualizations helped in understanding the dataset and evaluating the model's predictions.

12. MODEL TRAINING USING LINEAR REGRESSION

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value based on one or more input variables.

In this project, Linear Regression was used to predict sales based on advertising expenditure on TV, Radio, and Newspaper.

The dataset was divided into training and testing sets. The training data was used to train the model, while the testing data was used to evaluate its performance on unseen observations.

The model was created using the LinearRegression class from the Scikit-learn library. The fit() method was used to train the model on the training data.

The model learns the relationship between advertising expenditure and sales. After training, it can use the input advertising values to estimate sales.

The general equation of multiple linear regression is:

$Sales = b_0 + b_1(TV) + b_2(Radio) + b_3(Newspaper)$

Where:

- **Sales:** Predicted sales value.
- **b_0:** Intercept of the model.
- **b_1, b_2, b_3:** Coefficients learned by the model.
- **TV, Radio, Newspaper:** Input features representing advertising expenditure.

After training, the model was ready to generate predictions for the testing dataset and new advertising inputs.

13. MODEL EVALUATION

Model evaluation is an important step in machine learning. It helps determine how well a trained model performs on testing data.

In this project, the Linear Regression model was evaluated using the following performance metrics:

1. Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual sales values and predicted sales values. A lower MAE indicates that predictions are closer to actual values on average.

2. Mean Squared Error (MSE)

MSE calculates the average of the squared differences between actual and predicted sales values. It gives greater weight to larger prediction errors. A lower MSE generally indicates better performance.

3. Root Mean Squared Error (RMSE)

RMSE is the square root of MSE. It measures prediction error in the same units as the target variable. A lower RMSE generally indicates that the model's predictions are closer to actual sales values.

4. R-Squared Score (R^2)

The R^2 score indicates how much of the variation in the target variable is explained by the model on the evaluated data. A value closer to 1 generally indicates a better fit.

These metrics were calculated using the Scikit-learn library to assess the performance of the Linear Regression model.

14. RESULTS AND DISCUSSION

After training the Linear Regression model, predictions were generated for the testing dataset. The predicted values were compared with the actual sales values.

The model's performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared score.

Performance Results

- Mean Absolute Error (MAE): [Enter your Colab output]
- Mean Squared Error (MSE): [Enter your Colab output]
- Root Mean Squared Error (RMSE): [Enter your Colab output]
- R-squared Score (R^2): [Enter your Colab output]

The actual-versus-predicted scatter plot was also examined to understand how closely the predicted values matched the actual values.

The results help evaluate the usefulness of Linear Regression for predicting sales from advertising expenditure. The model's performance should be interpreted using the calculated metrics rather than assuming a particular accuracy in advance.

15. NEW SALES PREDICTION

After training and evaluating the Linear Regression model, it was used to predict sales for a new set of advertising expenditure values.

The model received the following input values:

- TV Advertising: 150
- Radio Advertising: 25
- Newspaper Advertising: 20

These values were provided to the trained model to generate a predicted sales value.

The prediction demonstrates how a trained machine learning model can estimate sales based on advertising expenditure. The predicted value depends on the relationships learned from the training dataset.

This feature can help businesses explore possible sales outcomes for different advertising budgets. However, the prediction is an estimate and should not be treated as a guarantee of actual sales.

16. CONCLUSION

The Sales Prediction Using Python project demonstrates the application of machine learning for estimating sales based on advertising expenditure.

The Advertising dataset was explored and visualized using Pandas, Matplotlib, and Seaborn. A Linear Regression model was developed using Scikit-learn, with TV, Radio, and Newspaper advertising expenditure as input features and Sales as the target variable.

The model was trained on the training dataset and evaluated on the testing dataset using MAE, MSE, RMSE, and the R-squared score.

The project provided practical experience in data loading, data exploration, visualization, model training, prediction, and performance evaluation.

Overall, this project illustrates how Python and machine learning techniques can support data-driven decision-making in sales planning.

17. FUTURE SCOPE

The project can be improved and extended in several ways:

1. **Use More Data:** A larger dataset containing more observations may help improve the model's usefulness.
2. **Include Additional Features:** Factors such as product price, seasonal trends, customer demand, and promotional offers could be considered if suitable data is available.
3. **Compare Different Algorithms:** Models such as Decision Tree Regression, Random Forest Regression, and other regression techniques could be compared with Linear Regression.
4. **Improve Model Evaluation:** Cross-validation and additional evaluation techniques could be used to assess model performance more reliably.
5. **Develop a User Interface:** A simple web application could allow users to enter advertising expenditure and view predicted sales.
6. **Support Business Planning:** With suitable real-world data and further testing, the model could be explored as one input for advertising and sales planning.

These improvements could make the project more useful for practical sales forecasting applications.

18. REFERENCES

The following resources were used as references while developing this project:

1. **Pandas Documentation** — https://pandas.pydata.org/docs/
2. **NumPy Documentation** — https://numpy.org/doc/
3. **Matplotlib Documentation** — https://matplotlib.org/stable/
4. **Seaborn Documentation** — https://seaborn.pydata.org/
5. **Scikit-learn Documentation** — https://scikit-learn.org/stable/
6. **Google Colab** — https://colab.research.google.com/

These resources provide documentation and guidance on data analysis, visualization, machine learning, and Python programming.